

Logical Schema

Users Entity: Represents all system actors (Tourists, Guides, Admins) with mandatory Email uniqueness

Guides Entity: A specialized extension of the User entity (1:1 Relationship) requiring Admin approval and official licensing

Destinations Entity: Central repository for all Egyptian touristic landmarks categorized by region

Booking Entities: Divided into **Trip Bookings** (General site visits) and **Guide Bookings** (Personalized tours)

Reviews Entity: Feedback mechanism restricted to a 1:1 relationship with completed bookings to ensure data integrity

Relational Logic: Guides are linked to multiple languages and regions to enable advanced filtering for tourists

Physical Schema

Table Name	Data Types & Storage	Constraints & Optimization
Users	INT IDENTITY, NVARCHAR(150), BIT	Primary Key, Unique .Constraints, Defaults
Guides	INT, NVARCHAR(50), NVARCHAR(MAX)	Foreign Key (UserId), .Check Constraints (Status)
Destinations	INT IDENTITY, NVARCHAR(500) for Images	Logical separation by .Region (Index optimized)
TripBookings	DATE, INT, NVARCHAR(20) for Status	Check (TripType), Foreign .Keys (Tourist, Destination)
GuideBookings	DATE, INT for duration, NVARCHAR	Cascading Foreign Keys, Update/Creation .timestamps
Reviews	TINYINT for Rating (1-5 range)	Unique Index on BookingId to prevent .duplicates
Indexes	B-Tree Indexing on Status & Foreign Keys	Performance optimization .for high-traffic queries